

NETWORK TRAFFIC ANALYSIS REPORT

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Date: April 20, 2026

Tool: Wireshark 4.6.4

1. OBJECTIVE: Capture and analyze live network traffic to identify protocols, classify traffic types, and document security observations.

2. FINDINGS

DNS - 422 packets captured. Domains queried in plain text, including grammarly.com, smallpdf.com, google.com, and github.com. Security note: DNS is unencrypted - destination intent is visible even when HTTPS is used.

HTTP - 6 packets captured, plain text traffic to neverssl.com fully visible, including GET requests, browser version, and operating system. Security note: Any attacker on the same network could read this data completely.

TLS/HTTPS - 2,235 packets captured. Traffic to encrypted sites is fully unreadable. Destination IP visible but payload protected. Security note: TLS correctly protects data in transit.

External Traffic - 1,055 packets Traffic originating from outside the 192.168.1.0/24 network range, including responses from CDN and cloud infrastructure.

3. SECURITY OBSERVATIONS

- DNS queries reveal browsing intent even on HTTPS connections - monitor DNS for threat hunting
- Any HTTP traffic on an enterprise network should trigger an alert
- High volume of TLS traffic to unknown IPs warrants investigation in a SOC environment

4. CONCLUSION Successfully captured 12,637 packets and applied targeted filters to isolate security-relevant traffic. Lab demonstrates core network analysis skills applicable to log management and threat detection roles.